

Name: _____

RATIO AND PROPORTION, (SCALE FACTORS)

Date: _____

LO: I can use scale factors to enlarge or reduce shapes and interpret map scales.

Scale factors

A **scale factor** is the multiplier that converts an original length to a new length. A scale factor greater than 1 **enlarges**; between 0 and 1 it reduces.

Maps and models: A scale of 1 : 50 000 means 1 cm on the map represents 50 000 cm (500 m) in real life.

Quick examples

●	SF = 3 means new = original $\times$ 3	enlargement
●	SF = $\frac{1}{2}$ means new = original $\times \frac{1}{2}$	reduction
●	SF = new length $\div$ original length	finding the scale factor
●	SF = 2 doubles the area	area scales by SF ² = 4

Key idea

Scale factor = new length divided by original length. Multiply every length by the scale factor.

$$\text{SF} = \frac{\text{new length}}{\text{original length}} \div$$

All lengths scale by the same factor

MODELLED EXAMPLES

Study each example carefully before attempting the practice questions.

Example 1 - finding a scale factor

A shape has side 4 cm. The enlargement has side 12 cm. Find the scale factor.

$$\text{SF} = \text{new} \div \text{original}$$

$$\text{SF} = 12 \div 4$$

$$\text{SF} = 3$$

$$\text{SF} = 3$$

Divide the new length by the original length.

Example 2 - using a scale factor

Enlarge a triangle with sides 3, 4, 5 cm by scale factor 2.

Multiply each side by 2

$$3 \times 2 = 6, 4 \times 2 = 8, 5 \times 2 = 10$$

New sides: 6, 8, 10 cm

6, 8, 10 cm

Every side is multiplied by the same scale factor.

Example 3 - fractional scale factor

Reduce a rectangle 12 cm by 8 cm using SF = $\frac{1}{4}$.

$$\text{Length: } 12 \times \frac{1}{4} = 3 \text{ cm}$$

$$\text{Width: } 8 \times \frac{1}{4} = 2 \text{ cm}$$

New rectangle: 3 cm by 2 cm

3 cm by 2 cm

Fractional scale factors make shapes smaller.

Example 4 - map scales

A map scale is 1 : 25 000. Two towns are 6 cm apart on the map. Find the real distance.

$$\text{Real distance} = 6 \times 25\,000 = 150\,000 \text{ cm}$$

$$\text{Convert: } 150\,000 \text{ cm} = 1500 \text{ m} = 1.5 \text{ km}$$

1.5 km

Map distance $\times$ scale = real distance. Then convert units.

YOUR TURN – PRACTICE (deliberate practice)

1. Finding scale factors

Find the scale factor from original to image.

- a) Original 5 cm, image 15 cm
- b) Original 8 cm, image 2 cm
- c) Original 6 cm, image 18 cm
- d) Original 10 cm, image 4 cm

2. Applying scale factors

Find the new length after enlargement.

- a) Length 7 cm, SF = 3
- b) Length 12 cm, SF = $\frac{1}{3}$
- c) Length 9 cm, SF = 4
- d) Length 20 cm, SF = $\frac{1}{5}$

3. Reverse problems

Find the original length.

- a) New length 24 cm, SF = 4. Original?
- b) New length 5 cm, SF = $\frac{1}{3}$. Original?
- c) New length 30 cm, SF = 6. Original?
- d) New length 3 cm, SF = $\frac{1}{5}$. Original?

4. Map scales

Use the map scale to find real distances.

- a) Scale 1 : 50 000. Map distance 4 cm. Real distance?
- b) Scale 1 : 25 000. Map distance 8 cm. Real distance?
- c) Scale 1 : 100 000. Real distance 5 km. Map distance?
- d) Scale 1 : 20 000. Map distance 12 cm. Real distance?

COMMON MISCONCEPTIONS – SPOT THE MISTAKE

Each statement shows a student's answer. Tick () the ones that are correct. If it is wrong, write the correct answer.

a) Scale factor = new length ÷ original length ☐

Correct answer: _____

Reason: _____

b) A scale factor of 2 doubles the area ☐

Correct answer: _____

Reason: _____

c) A scale factor between 0 and 1 reduces the shape ☐

Correct answer: _____

Reason: _____

d) 1 : 50 000 means 1 cm on map = 50 000 cm in real life ☐

Correct answer: _____

Reason: _____

e) A scale factor less than 1 makes the shape bigger ☐

Correct answer: _____

Reason: _____

6. CHALLENGE – WORD PROBLEMS (apply your skills)

a) A rectangular garden is **8 m by 12 m**. An architect draws it with scale factor $\frac{1}{200}$. What are the dimensions on the plan in cm? If the garden area in real life is turfed at **£5 per m²**, find the total cost.

Expression: _____

Simplified: _____

b) Two similar triangles have areas **9 cm²** and **81 cm²**. The smaller triangle has a base of **6 cm**. Find the scale factor of enlargement and the base of the larger triangle.

Expression: _____

Simplified: _____

TOP TIPS

Read the question carefully. Show all your working step by step.
Check your answer makes sense.

CHECK YOUR WORK

Have I shown all my working clearly?

Did I check my answer using a different method?

Does my answer look reasonable?

